

A C++ based driver model.

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Presentations

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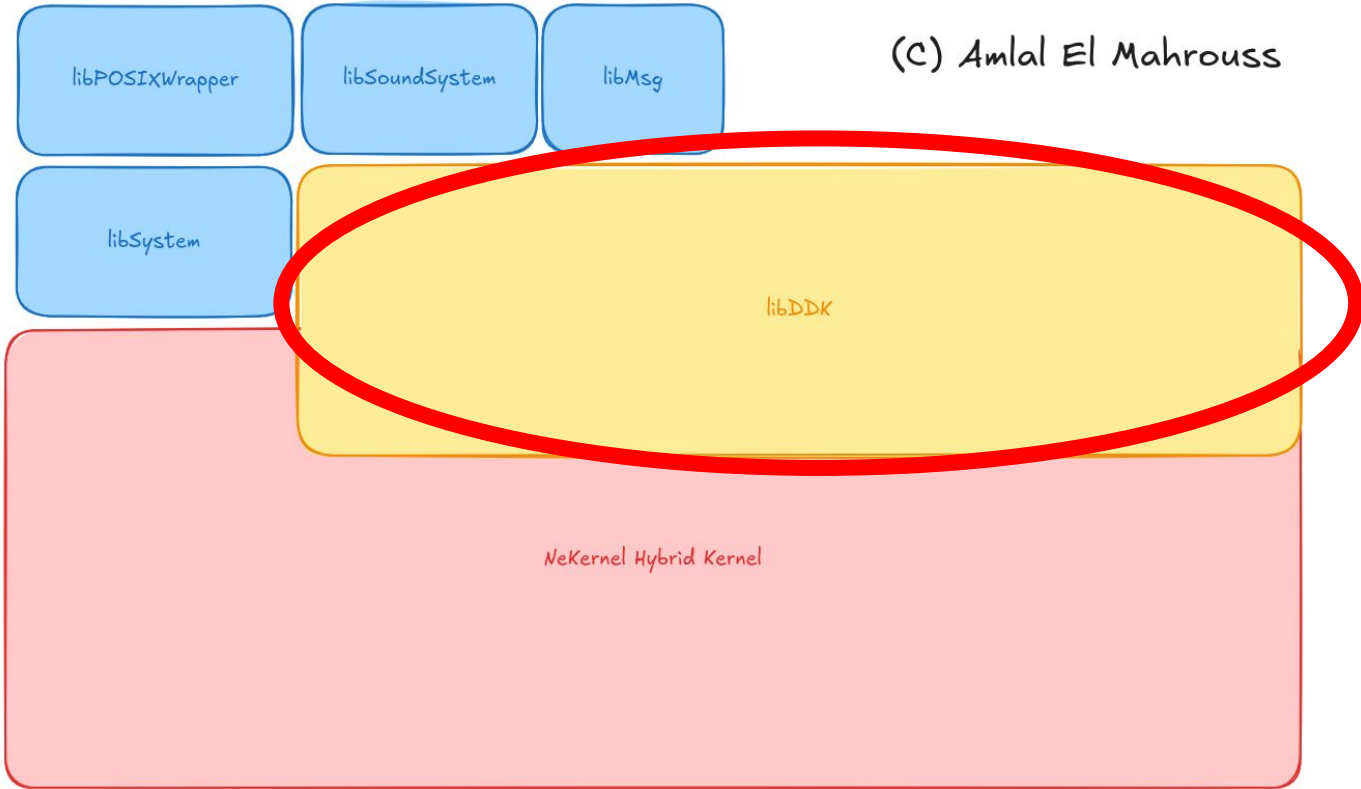
AMLAL EL MAHROUSS



- C++ Software Developer.
- Loves C++ and Systems Engineering.
- Loves Free Software.

The Architecture:

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The Design:

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The Rationale:

- It is based on C++17 and later.
- It sits between the kernel and userspace libraries/programs.
- It is ABI-agnostic because of the dispatch system.

```
/**
 * \brief Allocates a new heap on the Kernel's side.
 * \param sz the size of the heap block.
 * \return the newly allocated pointer.
 */
DDK_EXTERN void* kalloc(size_t sz) {
    if (!sz) ++sz;

    void* ptr = ::ke_call_dispatch("mm_alloc_ptr" 1, &sz, sizeof(size_t));
```

```
/**
 * \brief Frees a pointer.
 * \param ptr the pointer to be freed.
 */
DDK_EXTERN void kfree(void* ptr) {
    if (!ptr) return;

    ::ke_call_dispatch("mm_free_ptr" 1, &ptr, sizeof(void*));
```

The Dispatch System:

```
/**
 \brief Allocates a new heap on the Kernel's side.
 \param sz the size of the heap block.
 \return the newly allocated pointer.
 */
DDK_EXTERN void* kalloc(size_t sz) {
    if (!sz) ++sz;

    void* ptr = ::ke_call_dispatch("mm_alloc_ptr", 1, &sz, sizeof(size_t));
    return ptr;
}
```


Thanks! 😊

The Code and website are linked below:

Repository: <https://git.src.nekernel.org/ne-kernel>

Site: <https://src.nekernel.org>